MEDIUM · ADVANCED MATH

Advanced Math

17

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

$$fx = 3{,}0000.75^x$$

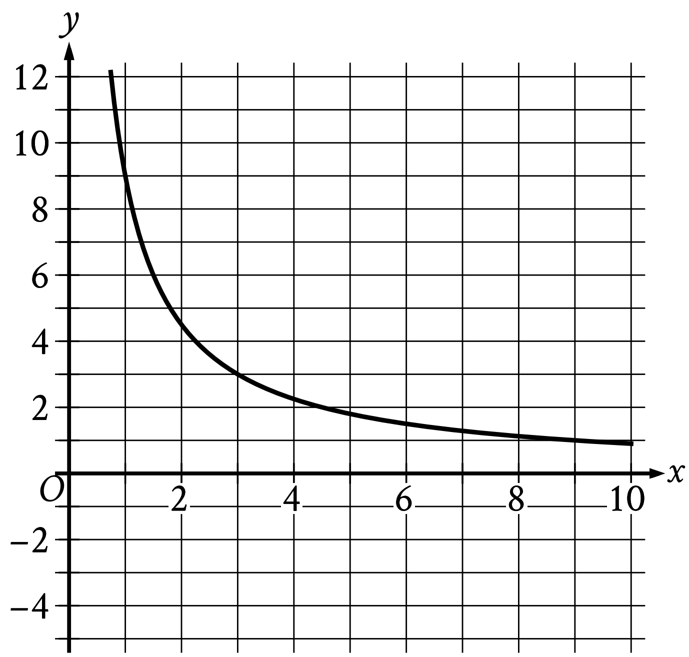
A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 3,000 in this context?

A. The estimated percent decrease in the population for this species and area every 8 years after 2008

B. The estimated percent decrease in the population for this species and area each year after 2008

C. The estimated population for this species and area 8 years after 2008

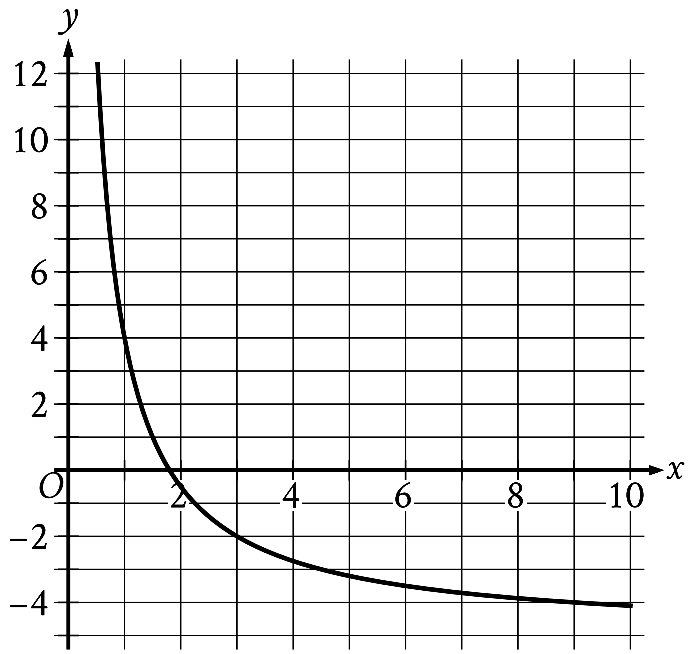
D. The estimated initial population for this species and area in 2008



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma nine halves)
  - (3 comma 3)
  - (6 comma three halves)

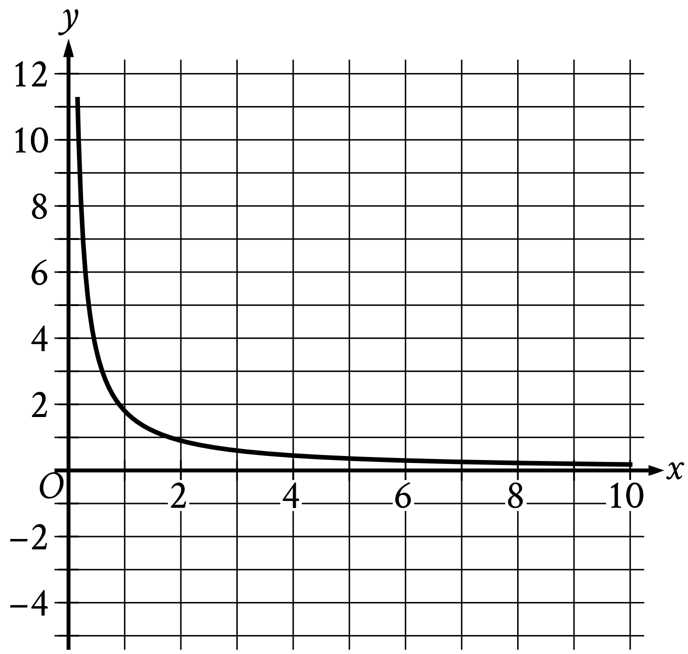
The graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. Which of the following is the graph of $y = f(x) + 5$, where $x \geq 0$?

A.



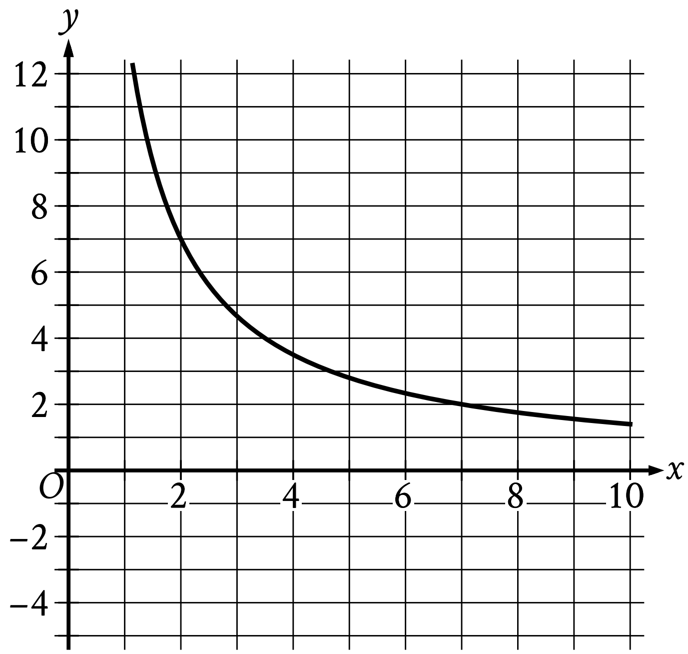
- Moving from left to right:
  - The curve passes from quadrant 1 to quadrant 4.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
  - (2 comma negative one half)
  - (3 comma negative 2)
  - (6 comma negative seven halves)

B.



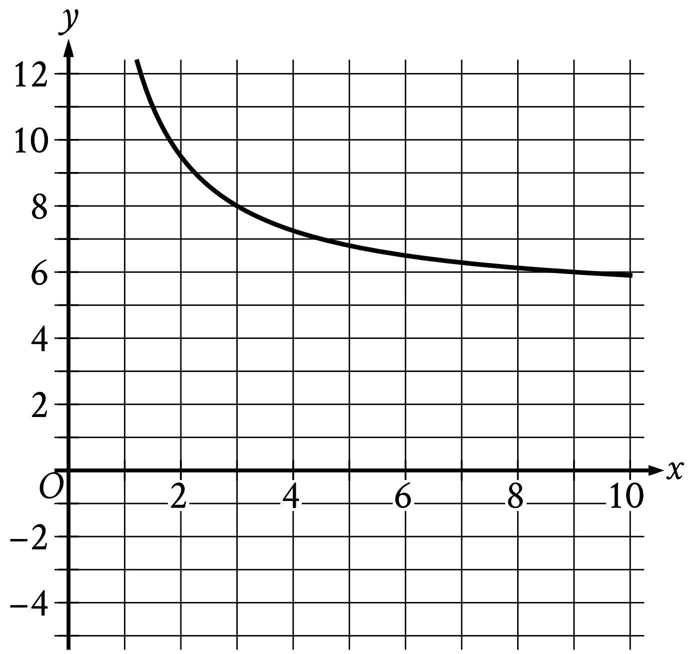
- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 1.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma nine tenths)
  - (3 comma three fifths)
  - (6 comma three tenths)

C.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 2.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
  - (2 comma 7)
  - (3 comma StartFraction 14 Over 3 EndFraction)
  - (6 comma seven thirds)

D.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to x equals 3.
  - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 5.
- The curve passes through the following points:
  - (2 comma StartFraction 19 Over 2 EndFraction)
  - (3 comma 8)
  - (6 comma StartFraction 13 Over 2 EndFraction)

Q3 ADVANCED MATH

There were no jackrabbits in Australia before 1788 when 24 jackrabbits were introduced. By 1920 the population of jackrabbits had reached 10 billion. If the population had grown exponentially, this would correspond to a 16.2% increase, on average, in the population each year. Which of the following functions best models the population $p(t)$ of jackrabbits t years after 1788?

A. $p(t) = 1.162(24)^t$

B. $p(t) = 24(2)^{1.162t}$

C. $p(t) = 24(1.162)^t$

D. $p(t) = (24 \cdot 1.162)^t$

Q4

GRID-IN ADVANCED MATH

$$y = 3{,}600a^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y, in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 ADVANCED MATH

$$ht = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 3,364 feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

A. 7.25

B. 14.50

C. 105.13

D. 210.25

Q6

ADVANCED MATH

A certain college had 3,000 students enrolled in 2015. The college predicts that after 2015, the number of students enrolled each year will be 2% less than the number of students enrolled the year before. Which of the following functions models the relationship between the number of students enrolled, $f(x)$, and the number of years after 2015, x ?

A. $f(x) = 0.02(3{,}000)^x$

B. $f(x) = 0.98(3{,}000)^x$

C. $f(x) = 3{,}000(0.02)^x$

D. $f(x) = 3{,}000(0.98)^x$

Q7 ADVANCED MATH

$$gx = 11\frac{1}{12}^{x}$$

If the given function g is graphed in the xy-plane, where $y = gx$, what is the y-intercept of the graph?

A. $0, 11$

B. $0, 132$

C. $0, 1$

D. $0, 12$

Q8 ADVANCED MATH

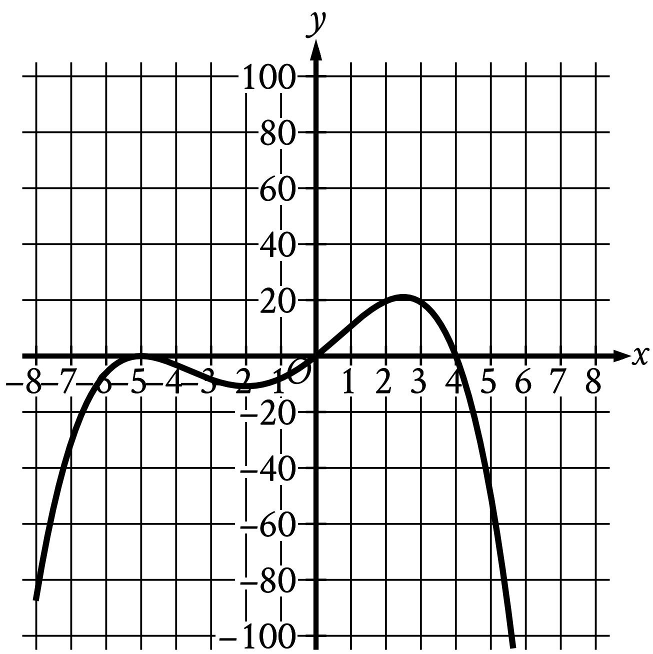
On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function f gives the predicted number of views x days after April 1. Which equation defines f?

A. $f x = 2330.70^{\frac{x}{2}}$

B. $f x = 2330.70^{2x}$

C. $f x = 2331.70^{\frac{x}{2}}$

D. $f x = 2331.70^{2x}$



- In quadrant 3:
  - The curve rises sharply to touch the x axis at point (negative 5 comma 0).
  - The curve falls gradually to a relative minimum at point (negative 2 comma negative 11).
  - The curve rises gradually to cross both axes at the origin.
- In quadrant 1:
  - The curve rises gradually to a relative maximum at point (2.5 comma 21).
  - The curve falls sharply to cross the x axis at point (4 comma 0).
- In quadrant 4 the curve falls sharply.

Which of the following could be the equation of the graph shown in the *xy*-plane?

A. $y = -\frac{1}{10}xx - 4x + 5$

B. $y = -\frac{1}{10}xx - 4x + 5^2$

C. $y = -\frac{1}{10}xx - 5x + 4$

D. $y = -\frac{1}{10}xx - 5^2x + 4$

Q10

ADVANCED MATH

A function p estimates that there were 2,000 animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where px is the estimated number of animals in the population x years after 1998?

A. $px = 2{,}0003^x$

B. $px = 2{,}0001.97^x$

C. $px = 2{,}0001.03^x$

D. $px = 2{,}0000.97^x$